

# LiteRehab Fusion

Real-time feedback for upper-limb rehabilitation practice at home.

## Summary

LiteRehab Fusion is a coursework engineering prototype that combines a forearm-worn motion sensor with camera-based posture estimation. The system recognises the demonstrated exercise, counts repetitions, checks selected movement features, and records synchronized motion and posture data. It is designed to support home practice under professional guidance; it does not make treatment decisions or assess clinical recovery.

## Background

Home-exercise programmes ask patients to practise away from the clinic. During these sessions, a patient may not notice that a movement is too fast, that the range is limited, or that the trunk is compensating. The physiotherapist usually reviews progress at a later appointment rather than observing each repetition.

## Aim

The project examines whether wearable motion sensing and a camera view of posture can provide immediate, understandable feedback during a prescribed upper-limb exercise. A synchronized session record is retained for possible later review.

## System design

### Wearable unit

An MPU6050 samples forearm motion at 50 Hz. An ESP32 wearable transmits the samples over BLE to an ESP32-S3 receiver, which maintains the local exercise state and repetition count.

### Posture input

A MaixCAM2 video stream supplies a wider view of the movement. Computer-based posture estimation is used to check trunk position alongside the forearm signal.

### Analysis and interface

The dashboard aligns the motion and posture streams, displays the current status, and stores a synchronized session record. A classroom CNN-BiGRU baseline is available with a rule-based fallback.

## Prototype capabilities

The current MVP demonstrates:

- elbow-flexion and forearm-rotation exercise recognition;
- repetition counting from the wearable motion signal;
- checks for excessive speed and insufficient movement range;
- a posture check for trunk compensation;
- local status feedback and synchronized IMU and pose logging.

## Intended use

### During home practice

The person receives feedback while performing the exercise, leaving time to adjust a later repetition in the same session.

### Between appointments

A physiotherapist could review the session record when discussing home practice. Exercise selection and all clinical decisions remain with the physiotherapist.

## Limitations and next steps

The exercise set is limited, and the classroom model was trained on a small public dataset. The prototype has not been clinically validated and makes no claim about diagnostic accuracy or patient outcomes.

The next step is supervised usability testing with physiotherapists and representative users. This work would examine the clarity and timing of the feedback and guide the collection of more representative movement data.